Practical No 03

a) Develop an Android application using LinearLayout (Horizontal and Vertical) to design structured UI.

Project Name: LinearLayout

Program Code:

Filename: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Student Registration"
        android:textSize="26sp"
        android:textStyle="bold"
        android:layout_gravity="center"
        android:layout_marginBottom="20dp"/>

    <EditText
        android:id="@+id/etName"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Name"
        android:layout_marginBottom="10dp"/>

    <EditText
        android:id="@+id/etEmail"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Email"
        android:layout_marginBottom="20dp"/>

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal"
        android:gravity="center">

        <Button
            android:id="@+id/btnSubmit"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Submit"/>

        <Button
            android:id="@+id/btnReset"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Reset"
            android:layout_marginStart="20dp"/>

    </LinearLayout>
</LinearLayout>
```

b) Develop an Android application using RelativeLayout to align UI components dynamically.

Project Name: RelativeLayout

Program Code:

Filename: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="16dp"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/titleText"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="20dp"
        android:text="RelativeLayout Demo"
        android:textAlignment="center"
        android:textSize="20sp"
        android:textStyle="bold"/>

    <ImageView
        android:id="@+id/imageView6"
        android:layout_width="268dp"
        android:layout_height="126dp"
        android:layout_below="@id/titleText"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="20dp"
        app:srcCompat="@drawable/zoro"/>

    <EditText
        android:id="@+id/editTextText2"
        android:layout_width="match_parent"
        android:layout_height="64dp"
        android:layout_below="@id/imageView6"
        android:layout_marginTop="20dp"
        android:ems="10"
        android:hint="Project Name"
        android:inputType="text"/>

    <Button
        android:id="@+id/button"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_below="@id/editTextText2"
        android:layout_alignParentStart="true"
        android:layout_marginTop="20dp"
        android:text="Sumbit"/>

    <Button
        android:id="@+id/button2"
        android:layout_width="95dp"
        android:layout_height="52dp"
        android:layout_below="@id/editTextText2"
        android:layout_alignParentEnd="true"
        android:layout_marginTop="20dp"
```

```
        android:text="Reset"/>
</RelativeLayout>
```

c) Develop an Android application using TableLayout to design a structured Login/Registration Form .

Project Name: TableLayout

Program Code:

Filename: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:fillViewport="true"
    tools:context=".MainActivity">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:padding="24dp">

        <TextView
            android:id="@+id/tvHeader"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_marginBottom="32dp"
            android:text="Registration Form"
            android:textColor="@android:color/black"
            android:textSize="28sp"
            android:textStyle="bold"/>

        <TableLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:stretchColumns="1">

            <TableRow
                android:layout_width="match_parent"
                android:layout_height="wrap_content">

                <TextView
                    android:layout_width="wrap_content"
                    android:layout_height="wrap_content"
                    android:layout_gravity="center_vertical"
                    android:paddingEnd="12dp"
                    android:text="Name:"
                    android:textColor="@android:color/black"
                    android:textSize="16sp"
                    android:textStyle="bold"/>

                <EditText
                    android:id="@+id/etName"
                    android:layout_width="match_parent"
                    android:layout_height="wrap_content"
                    android:hint="Enter Name"
                    android:inputType="textPersonName"/>
            </TableRow>

            <TableRow
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:layout_marginTop="8dp">
```

```
        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_gravity="center_vertical"
            android:paddingEnd="12dp"
            android:text="Email:"
            android:textColor="@android:color/black"
            android:textSize="16sp"
            android:textStyle="bold"/>

        <EditText
            android:id="@+id/etEmail"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="Enter Email"
            android:inputType="textEmailAddress"/>
    </TableRow>

    <TableRow
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="8dp">

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_gravity="center_vertical"
            android:paddingEnd="12dp"
            android:text="Password:"
            android:textColor="@android:color/black"
            android:textSize="16sp"
            android:textStyle="bold"/>

        <EditText
            android:id="@+id/etPassword"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="Enter Password"
            android:inputType="textPassword"/>
    </TableRow>

    <TableRow
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="24dp">

        <LinearLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:layout_span="2"
            android:orientation="horizontal">

            <Button
                android:id="@+id/btnRegister"
                android:layout_width="0dp"
                android:layout_height="wrap_content"
                android:layout_marginEnd="8dp"
                android:layout_weight="1"
                android:backgroundTint="#3F51B5"
                android:text="Register"
                android:textAllCaps="false"
                android:textColor="@android:color/white"
                app:cornerRadius="20dp"/>

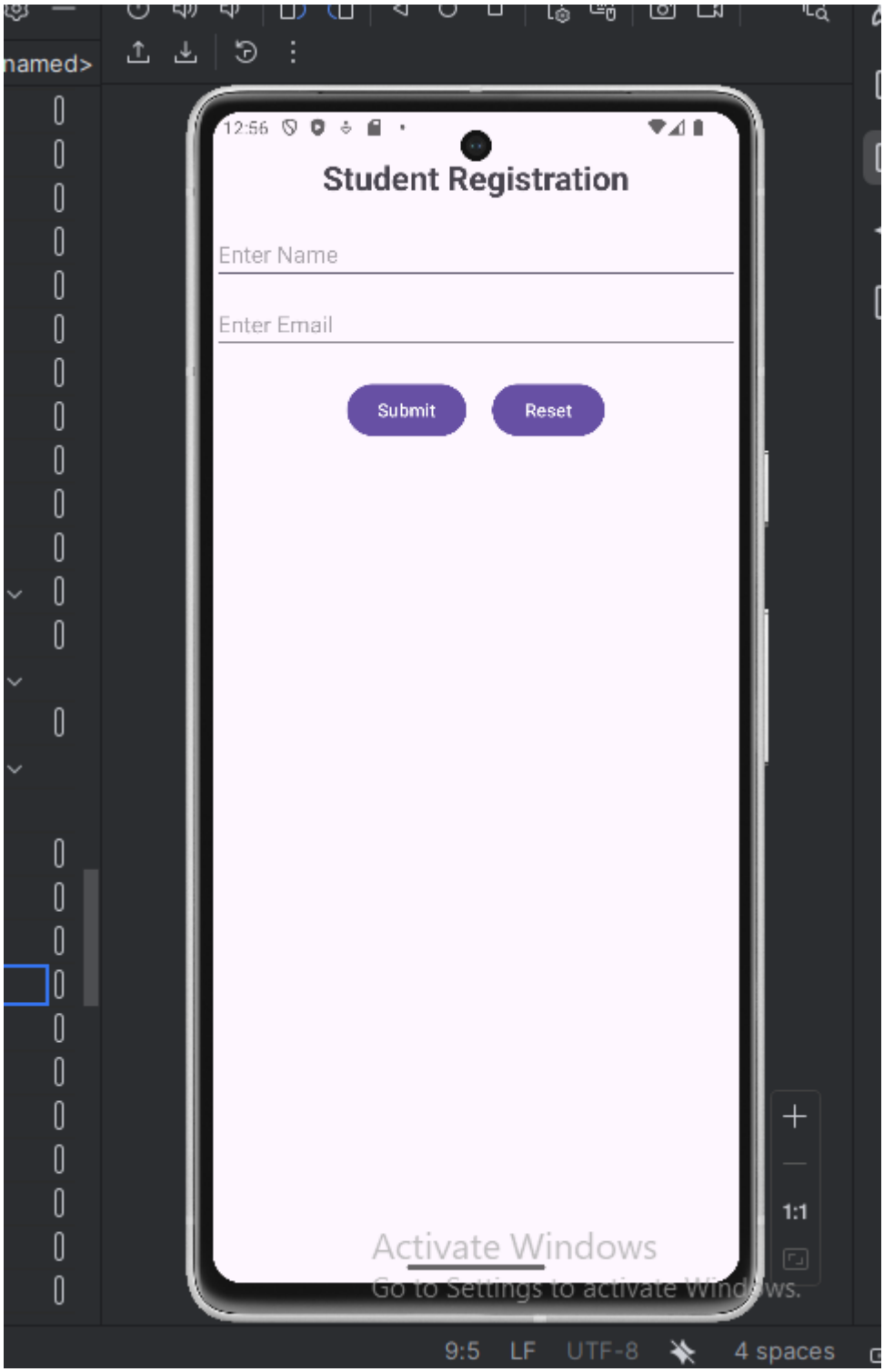
            <Button
                android:id="@+id/btnReset"
                android:layout_width="0dp"
```

```
                        android:layout_height="wrap_content"
                        android:layout_marginStart="8dp"
                        android:layout_weight="1"
                        android:backgroundTint="#3F51B5"
                        android:text="Reset"
                        android:textAllCaps="false"
                        android:textColor="@android:color/white"
                        app:cornerRadius="20dp"/>

                </LinearLayout>
            </TableRow>
        </TableLayout>
    </LinearLayout>
</ScrollView>
```

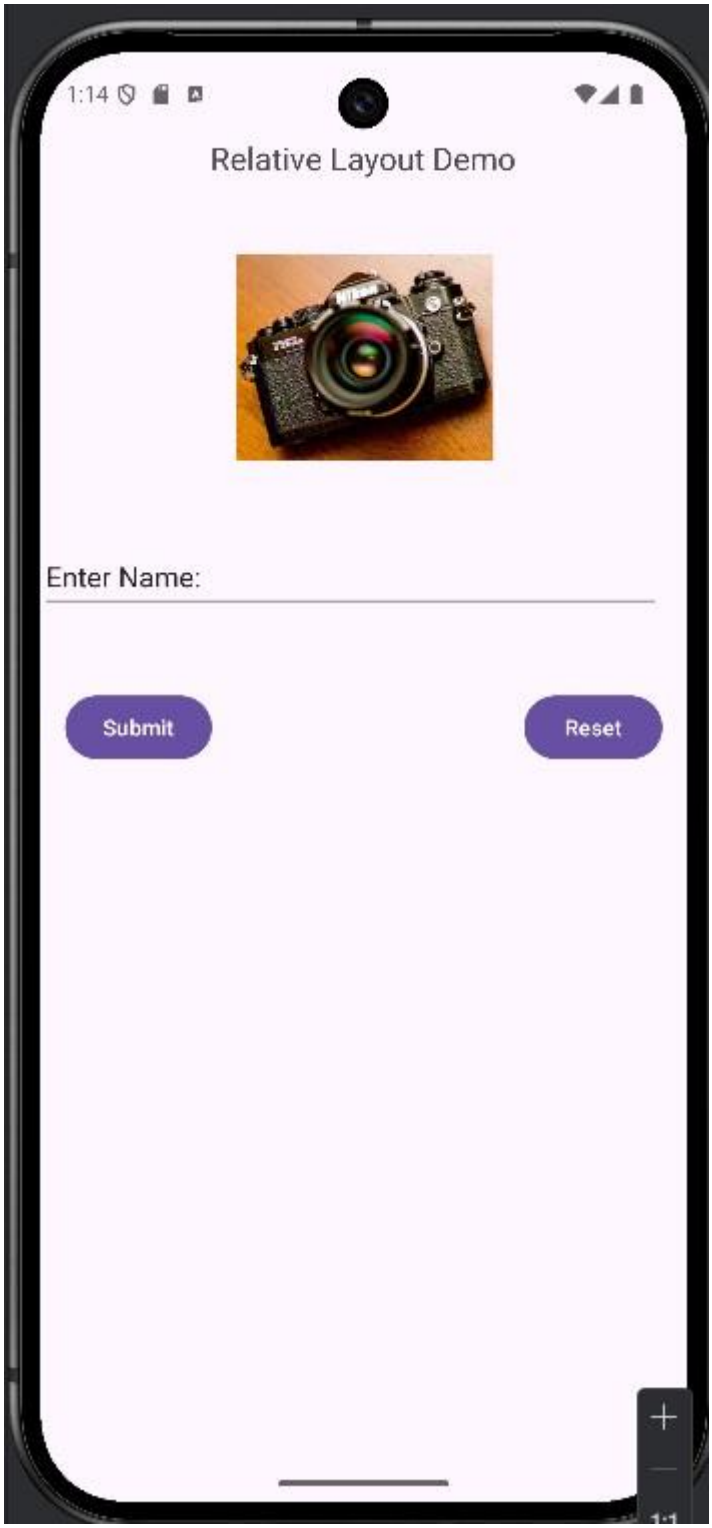
a) **Output:**

Linear Layout



b) <u>Output:</u>

Relative Layout



c) **Output:**

Table Layout

